

DonorHyperglycemiaStatsProject

Running Code

Age Hypothesis Test

Welch Two Sample t-test

```
data: r_age by delayed_fn
t = 1.6211, df = 210.42, p-value = 0.1065
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -0.4454789  4.5697784
sample estimates:
mean in group 0 mean in group 1
    54.37939      52.31724
```

The null hypothesis is that the mean ages of both groups are equal, while the alternative hypothesis is that the mean ages are not equal. The test statistic follows a t distribution with 210.42 degrees of freedom. The p-value is 0.1065, which is above the significance level of 0.05. Therefore, we fail to reject the null hypothesis of the mean ages being the same, regardless of whether or not there was delayed liver graft function.

Sex Hypothesis Test:

The null hypothesis is that there is no relationship between sex and Delayed Liver Graft Function. The alternative hypothesis is that there is a relationship.

Pearson's Chi-squared test with Yates' continuity correction

```
data:  table(DonorHyperglycemia$r_genderf, DonorHyperglycemia$delayed_fn)
X-squared = 0.41952, df = 1, p-value = 0.5172
```

The test statistic 0.41952 follows a Chi-squared distribution with one degree of freedom. The p-value of 0.5172 is greater than the significance level of 0.5, so we fail to reject the null hypothesis that there is no relationship. There is insufficient evidence to support there being a relationship between gender and delayed liver graft function.